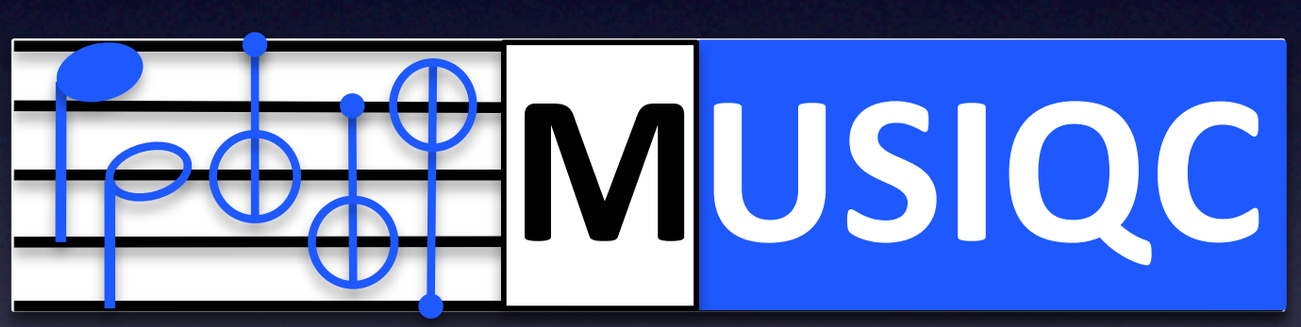
Quantum State and Process Tomography (and Quantum Algorithms)



Dave Bacon (University of Washington)

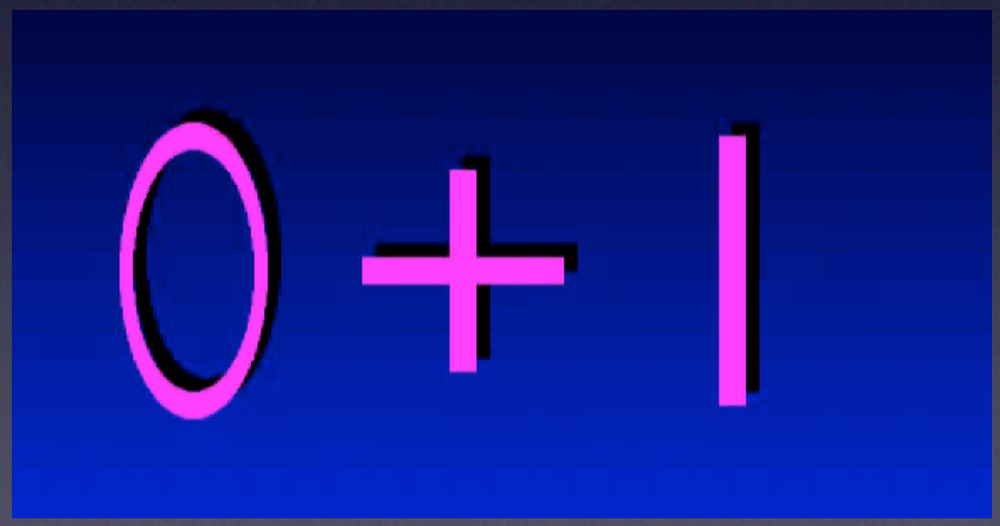
MUSIQC Kickoff Meeting

October 14, 2010

QW ("Q"-Dub) Theory Group

Dave Bacon
Aram Harrow

Gregory Crosswhite (Physics)
Paul Pham (CS)
Jijian Yan (Physics)
Lukas Svec (Physics)
Isaac Crosson (Physics)
Kamil Mitchnicki (Physics)
David Rosenbaum (CS)
Kevin Zatloukal (CS)



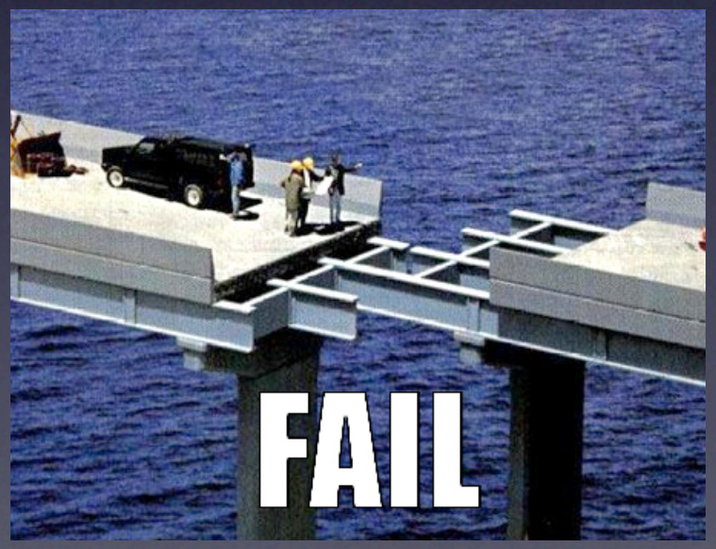
(this animation brought to you by a procrastinating grad student)

Overview

- To verify the workings of the MUSIQC prototypes, we must provide tools for characterizing the basic operating units acting alone and acting together.
- For small systems this is tractable, but for larger systems this quickly becomes intractable (time to perform >> time to obtain Ph.D.)
- However [insert evil theorist laugh] there are “ways” around this...

Goals

- Provide tools and protocols necessary for the experimentalists to perform tractable heralded ansatz tomography (state and process). Pass results of this onto system model / FT efforts.
- Design quantum algorithms that are specifically designed to make the above procedures fail.



[UW, UBC, Georgia Tech, Michigan, plus integration into experimental efforts]

Qubits Tomography

State: pure mixed

$|\psi\rangle$ ρ

$2^n - 2$ $4^n - 1$ ← parameters

Qubits Tomography

State:	pure	mixed		Process:	unitary	open
	$\|\psi\rangle$	ρ			U	$\$[\cdot]$
	$2^n - 2$	$4^n - 1$	← parameters →		$4^n - 1$	$8^n - 4^n$

Qubits Tomography

<u>State:</u>	pure	mixed		<u>Process:</u>	unitary	open
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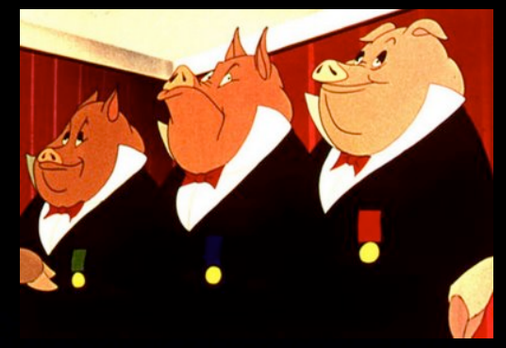
Subtleties:

- Estimating potentially very small numbers
- Given experimental data, some methods produce states or processes that are not physically allowed (Example: may produce negative probabilities!)
- Tomography process is noisy

Obstacles:

- Intractable!!!

Not All States Are Created Equal

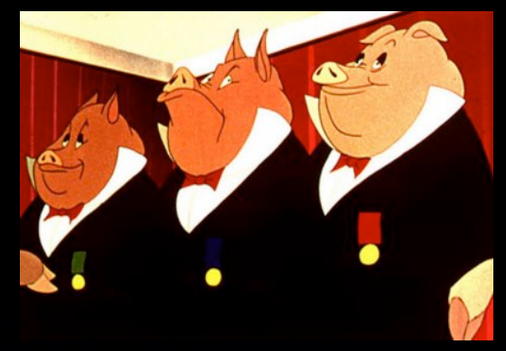


Separable states on *n* qubits

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle \qquad \rho_1 \otimes \rho_2 \otimes \cdots \otimes \rho_n$$

$2n$ ← parameters → $3n$

Not All States Are Created Equal



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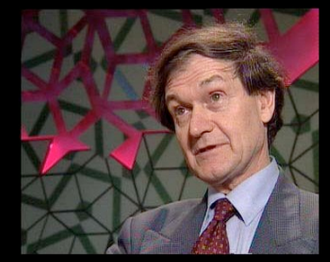
$2n$ ← parameters → $3n$

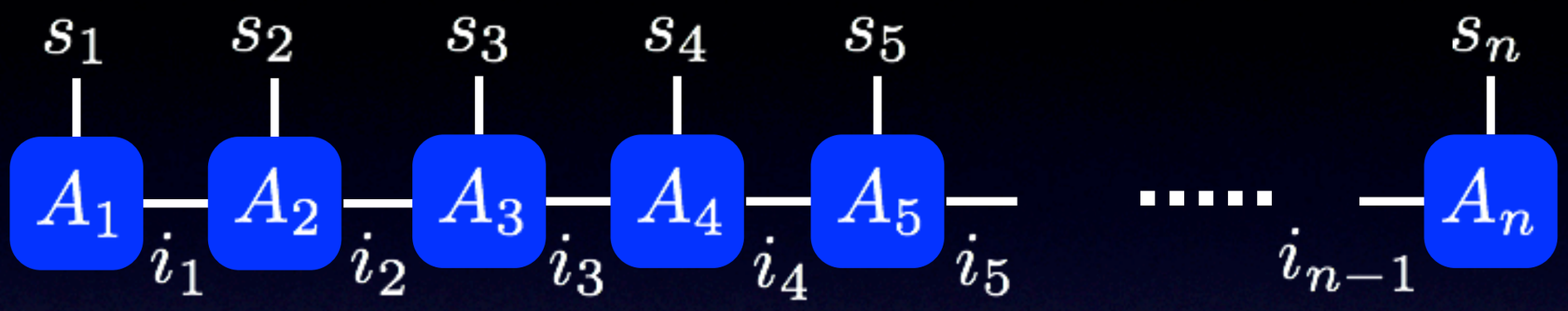
We would like to consider methods for

- Efficiently classically store information about state
- Use this information to efficiently calculate results of few qubit measurements
- Shouldn't have "holes": inefficient use can capture generic state

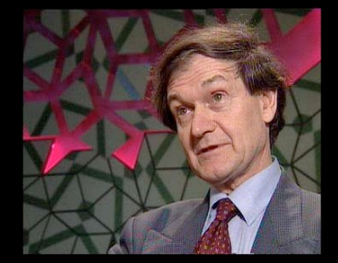
....data structure for quantum states

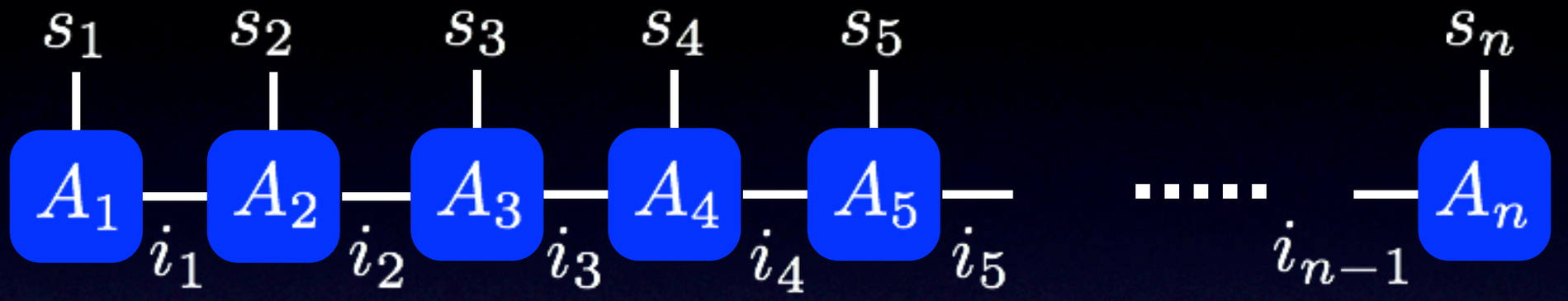
Matrix Product States





Matrix Product States



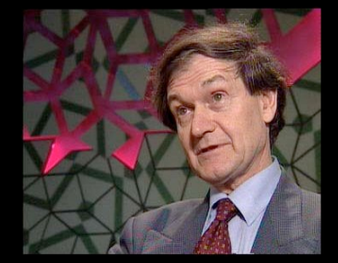


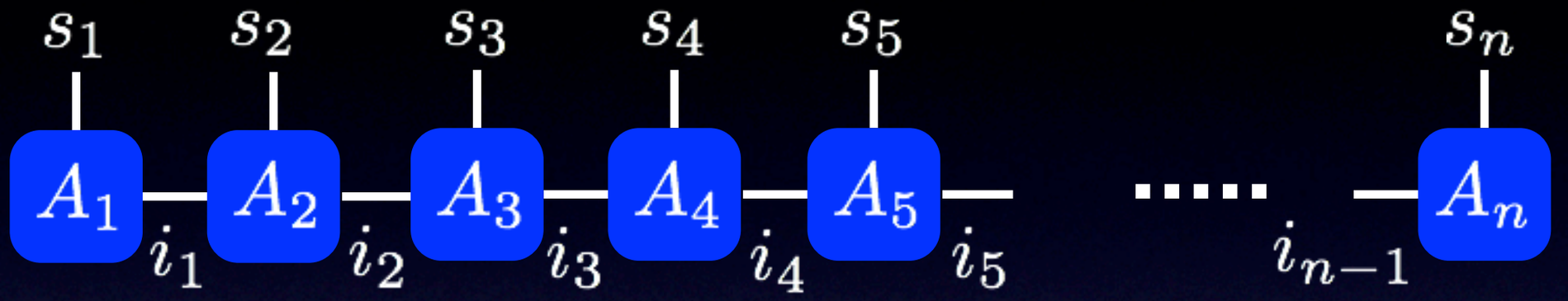
$$\sum_{s_1,\ldots,s_n \in \{0,1\}} \sum_{i_1,\ldots,i_n \in \{1,\ldots,D\}} [A_1]^{s_1}_{i_1} [A_2]^{s_2}_{i_1,i_2} [A_3]^{s_3}_{i_2,i_3} \cdots [A_n]^{s_n}_{i_{n-1}} |s_1,\ldots,s_n\rangle$$

$2D$ $2D^2$

total parameters $\longrightarrow$ $2(D^2(n-2)+2D) = O(D^2 n)$

Matrix Product States





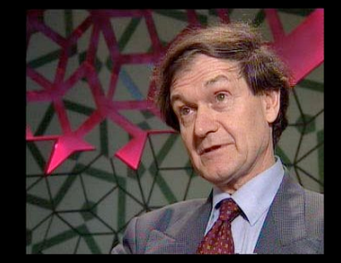
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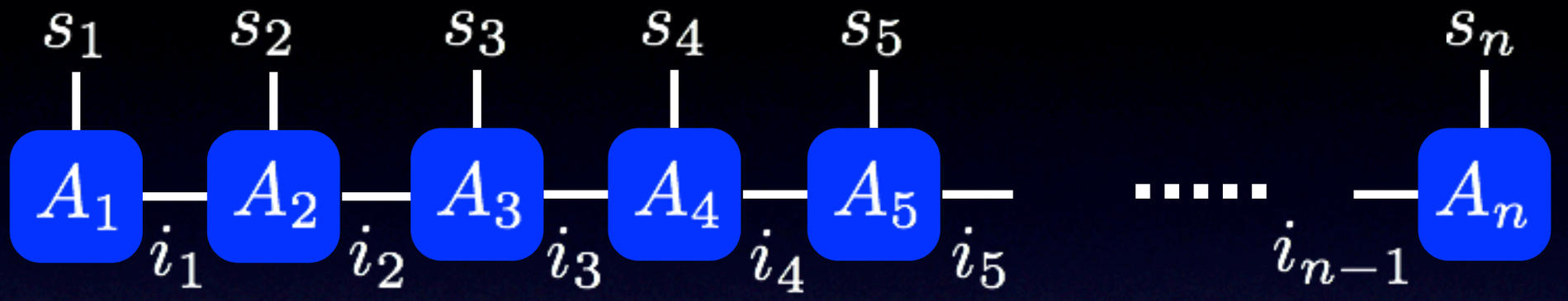
$2D$ $2D^2$

total parameters $\longrightarrow$ $2(D^2(n-2)+2D) = O(D^2 n)$

If *D* is polynomial in *n*, then # parameters poly in *n*

Matrix Product States





$$\sum_{s_1,\ldots,s_n \in \{0,1\}} \sum_{i_1,\ldots,i_n \in \{1,\ldots,D\}} [A_1]^{s_1}_{i_1} [A_2]^{s_2}_{i_1,i_2} [A_3]^{s_3}_{i_2,i_3} \cdots [A_n]^{s_n}_{i_{n-1}} |s_1,\ldots,s_n\rangle$$

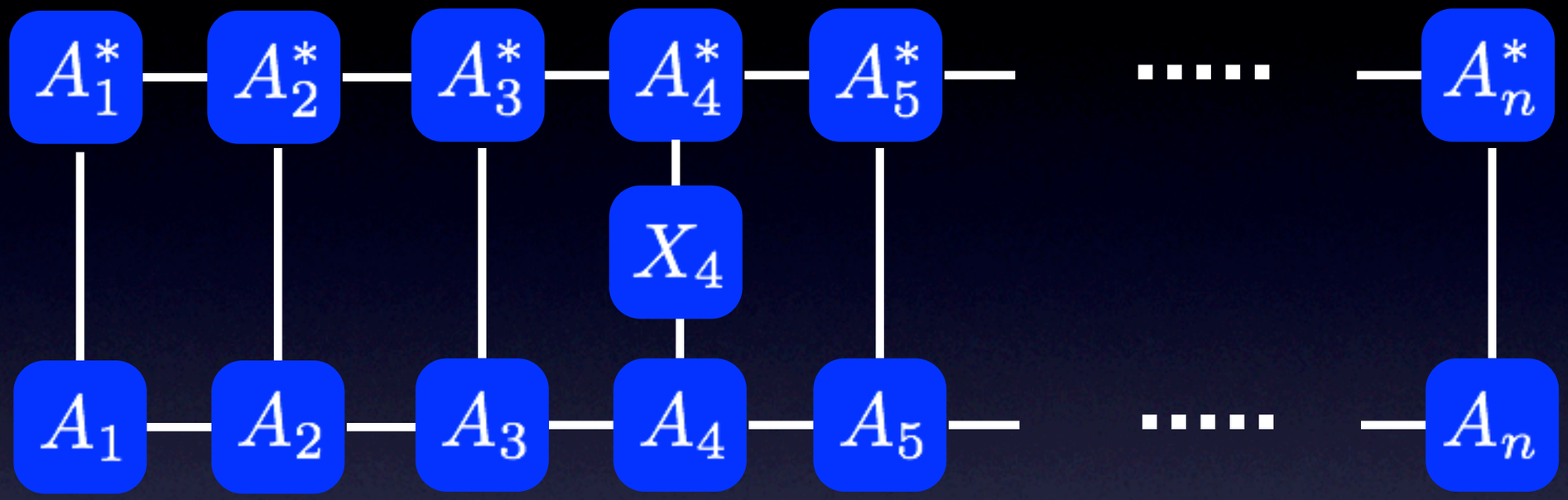
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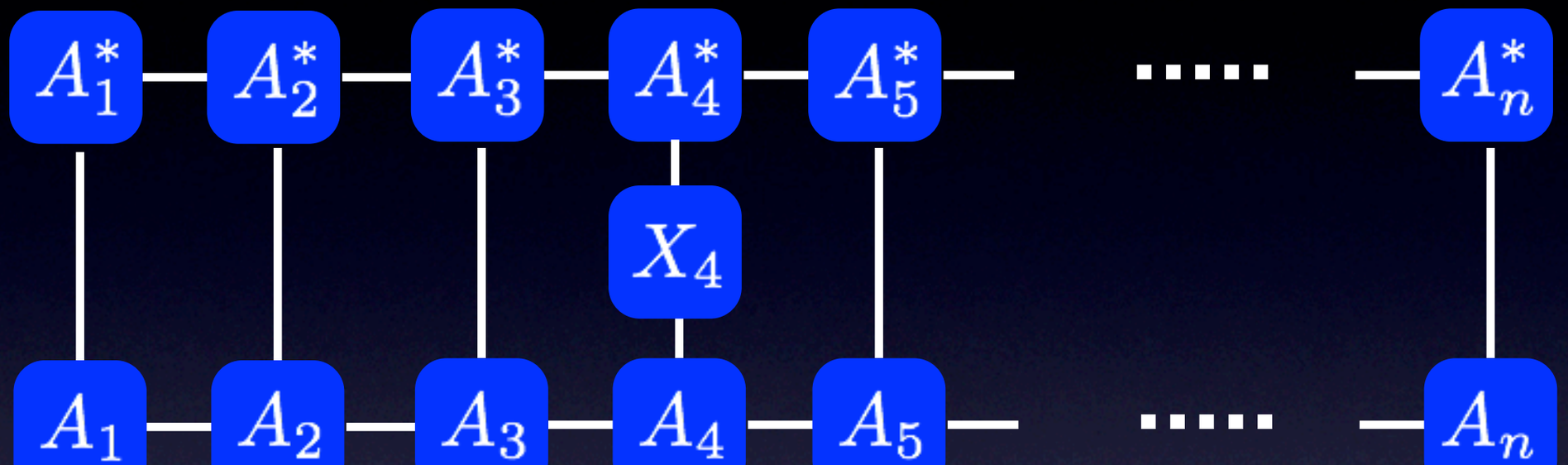
If *D* is polynomial in *n*, then # parameters poly in *n*

(D exponential in *n* captures all possible pure states)

Matrix Product States



Matrix Product States



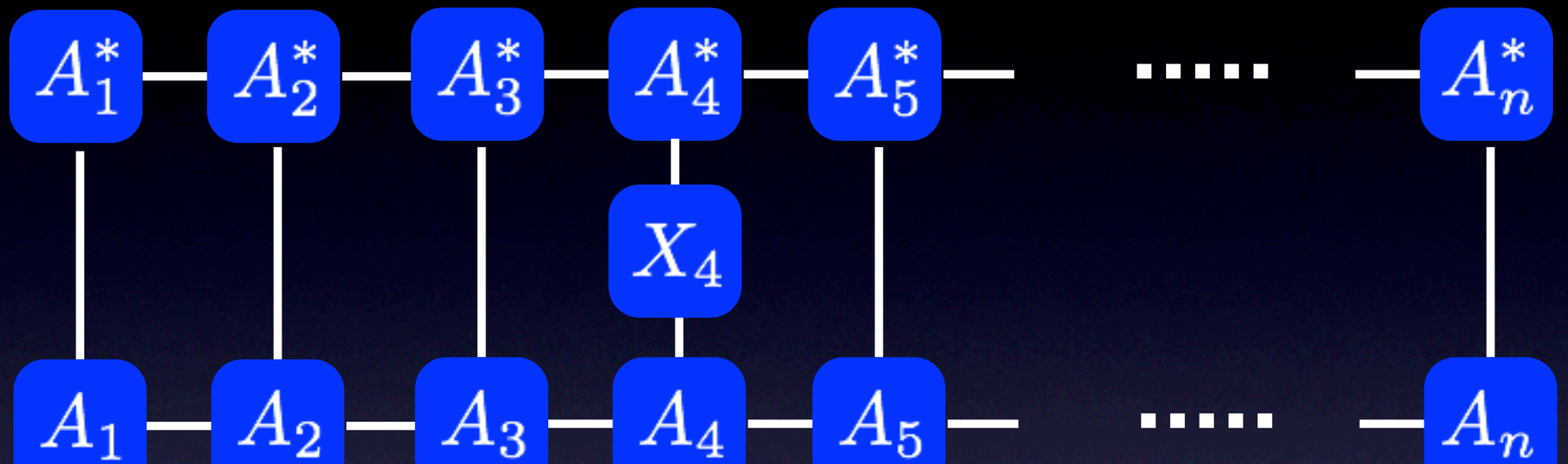
$$\langle X_4 \rangle =$$

$$\sum_{s'_1,\ldots,s'_n \in \{0,1\}} \sum_{i'_1,\ldots,i'_n \in \{1,\ldots,D\}} [A_1^*]^{s'_1}_{i'_1} [A_2^*]^{s'_2}_{i'_1,i'_2} [A_3^*]^{s'_3}_{i'_2,i'_3} \cdots [A_n^*]^{s'_n}_{i'_{n-1}} \langle s'_1, \ldots, s'_n |$$

$$\times X_4 \times$$

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Matrix Product States



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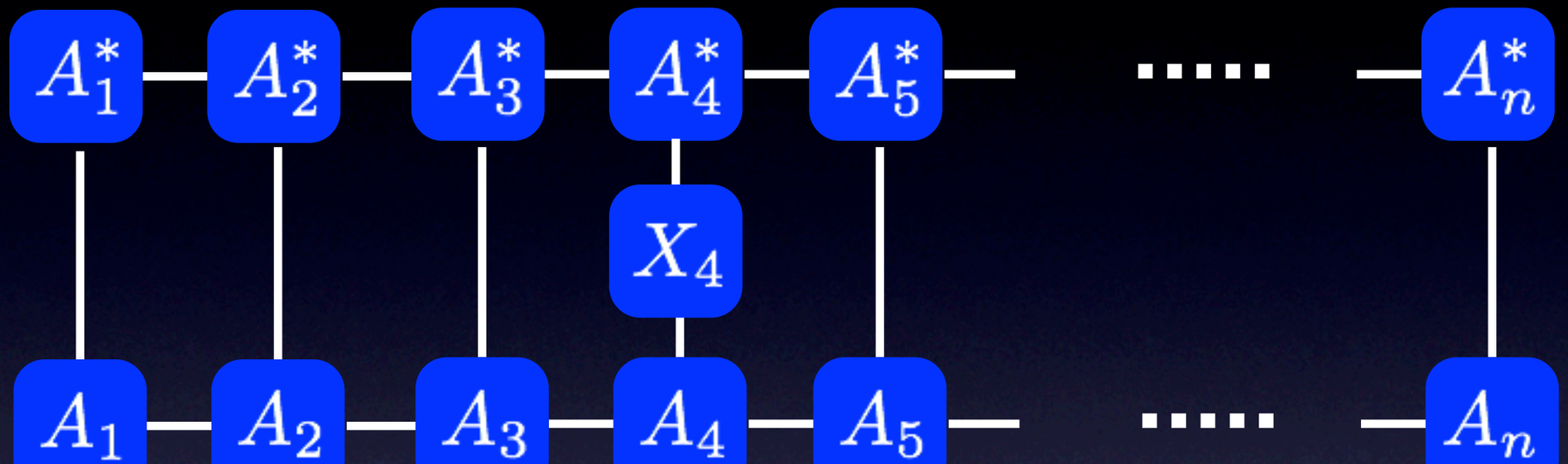
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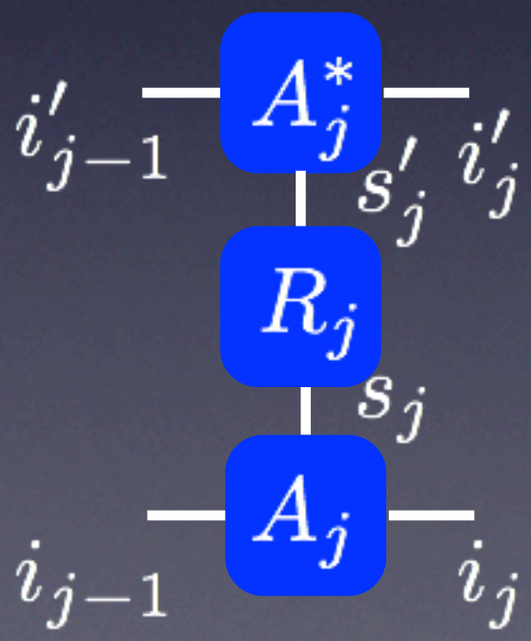
$$\times X_4 \times$$

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QMG

Matrix Product States

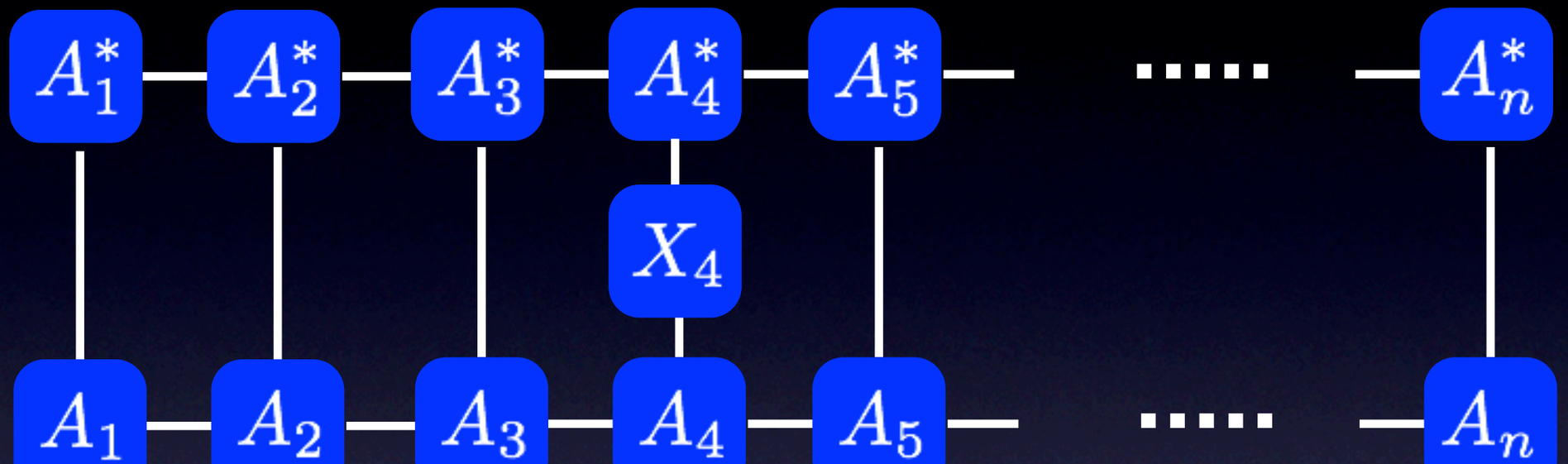


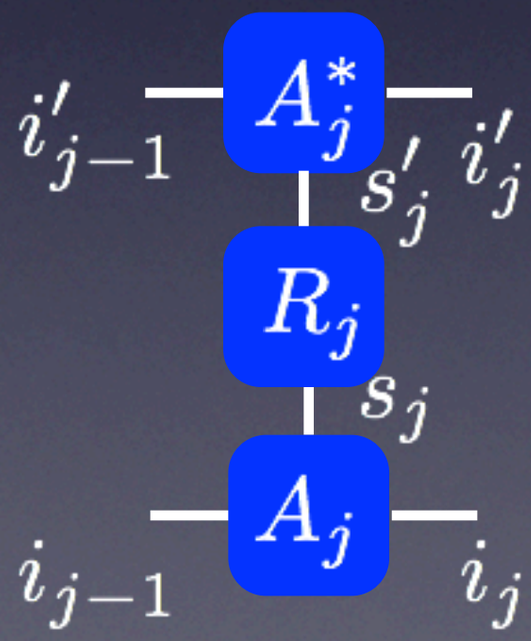


$$= \sum_{s_j, s'_j \in \{0,1\}} [A_j^*]^{s'_j}_{i'_{j-1}, i'_j} [R_j]_{s'_j, s_j} [A_j]^{s_j}_{i_{j-1}, i_j}$$

$$= [\tilde{R}_j]_{(i'_{j-1}, i_{j-1}), (i'_j, i_j)} =$$

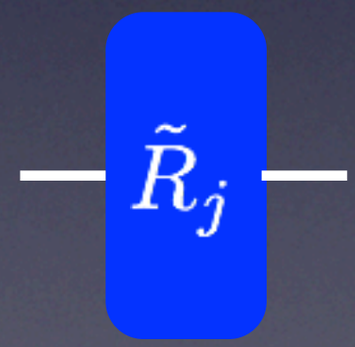
Matrix Product States





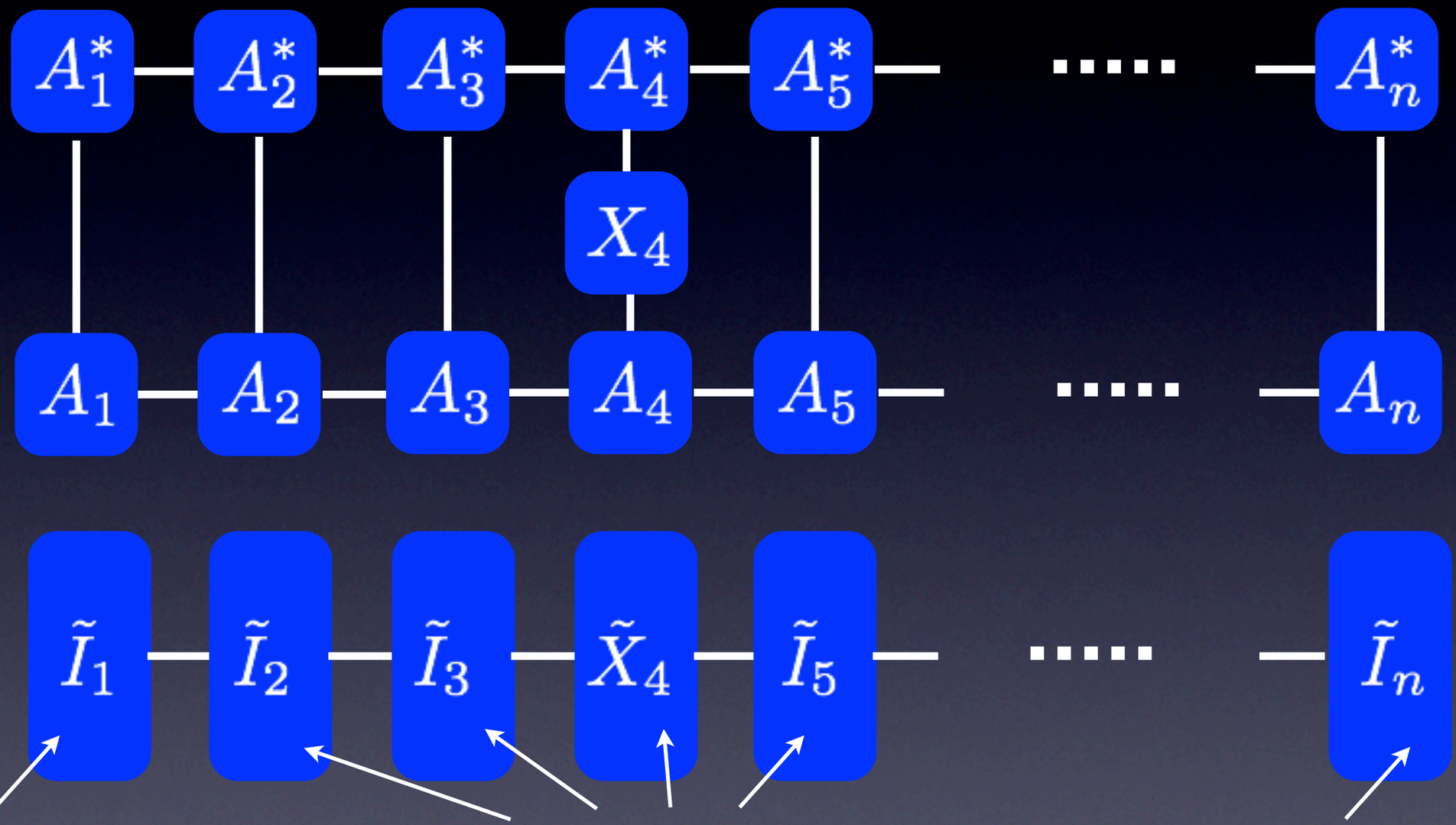
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$$= [\tilde{R}_j]_{(i'_{j-1}, i_{j-1}), (i'_j, i_j)} =$$



$D^2 \times D^2$ matrix

Matrix Product States



D^2-dim vector $D^2 \times D^2$ matrices D^2-dim vector

$$\langle X_4 \rangle = \langle \tilde{I}_1 | \tilde{I}_2 \tilde{I}_3 \tilde{X}_4 \tilde{I}_5 \cdots \tilde{I}_{n-1} | \tilde{I}_n \rangle$$

If *D* is polynomial in *n* then this calculation is polytime

Matrix Product States

✓ Efficiently classically store information about state

✓ Use this information to efficiently calculate results of few-qubit measurements

✓ Shouldn't have "holes": inefficient use can capture generic state

Matrix Product States

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MPS states are a useful ansatz:

- In practice capture one dimensional low energy eigenstates
- Can be used to simulate quantum algorithms beyond the tens of qubit regime
- Key to classical density matrix renormalization group algorithm

Matrix Product State Tomography

Can we make polynomial # measurement on n qubit state and learn a MPS description of the state (if it exists)?

"YES WE CAN"

[Cramer, Plenio arXiv:1002.3780]
[Flammia, Gross, Bartlett, Somma arXiv:1002.3839]
[Landon-Cardinal, Liu, Poulin arXiv:1002.4632]

MPS Pure State Tomography

Basic idea:

- Perform traditional (or compressed sensing based) tomography on constant k contiguous qubits



- Solve for MPS using variational method, DMRG based method, quantum 2-SAT approach
- Profit!

Matrix Product State Tomography

[Cramer, Plenio arXiv:1002.3780]

- Compressed sensing approach to obtaining local density matrices

[Flammia, Gross, Bartlett, Somma arXiv:1002.3839]

- Certificate

[Landon-Cardinal, Liu, Poulin arXiv:1002.4632]

- Works beyond "injective" MPS

All can be made to work for mixed states and process tomography by suitable hard work not carried out in these papers

Heralded Tomography

[Flammia, Gross, Bartlett, Somma arXiv:1002.3839]

By constructing the "parent Hamiltonian" of a MPS description one can use this to verify the accuracy of the MPS

$$\tau = \frac{1}{1-2\gamma}\sum_j (\mathrm{Tr}(h_j\sigma_j) + \epsilon_j)$$

"Announces when it fails"

Symmetrized Tomography

If states (processes) are symmetric with respect to some group, this can reduce the size of their description

Example: permutationally invariant qubit mixed states

[Toth, Wieczorek, Gross, Krischek, Schwemmer, Weinfurter arXiv:1005.3313]

Or we can enforce this symmetry, very useful for process tomography:

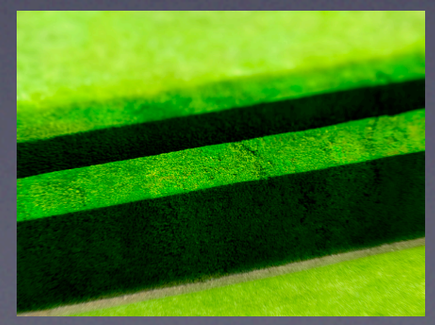
[Emerson, Silva, Moussa, Ryan, Laforest, Baugh, Cory, Laflamme, arXiv:0707.0685]

Tomography Tool Chest

Variety of tomography protocols: MPS, Symmetry based, Compressed sensing, MERA + other ansatz based methods

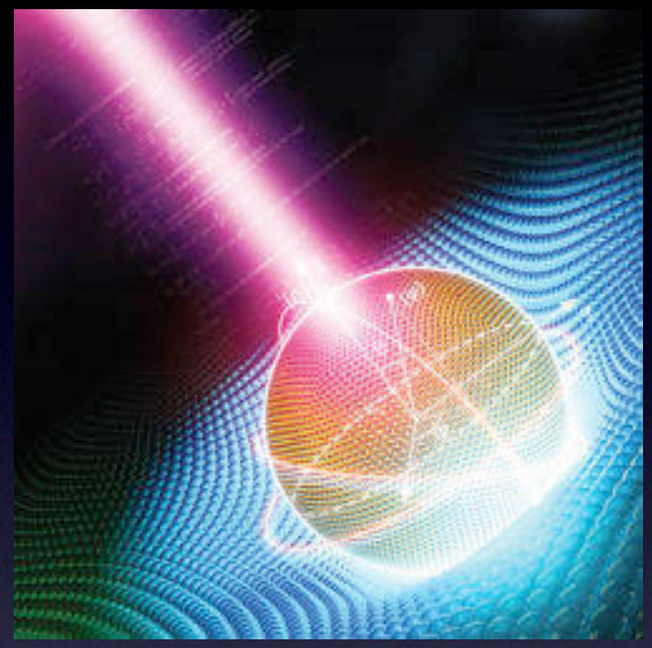
Estimation procedures: MPS and MERA (Variational, DMRG, Quantum 2-SAT), Maximum likelihood, Hedged maximum likelihood, Baysian methods(?)

Constraints: limits imposed by time constraints, # parallel operations, fidelity of operations



Quantum Algorithms

- Factoring
- Discrete Log
- Grover Search
- Amplitude Amplification
- Order Finding
- Solving Pell's Equation
- Heisenberg Hidden Subgroup
- Sampling Solutions to Linear Equations
- NAND Tree
- Quantum Simulation
- Quantum Error Correction
- .....



Which to choose?

Quantum Algorithms

Which to choose?

- System optimization study will identify what reasonable fault-tolerant quantum error correction plus algorithm can be implemented on MUSIQC
- In the mean time....

Starting a War With DMRG

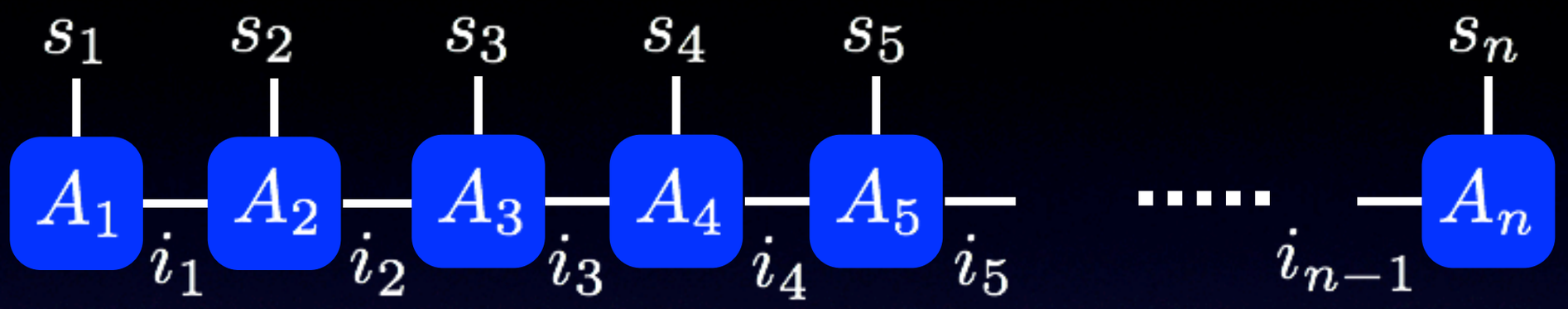
MUSIQC-80 will be in a size range where

- Simulating it using the best classical methods will start to be come difficult

Density matrix renormalization group works when there is a efficient MPS description (i.e. when our MPS tomography works)

Design efficient MUSIQC executable quantum circuits which have large MPS descriptions and challenge classical DMRG to simulate circuit

Why MPS classically efficient?



$$\sum_{s_1,\ldots,s_n \in \{0,1\}} \sum_{i_1,\ldots,i_n \in \{1,\ldots,D\}} [A_1]^{s_1}_{i_1} [A_2]^{s_2}_{i_1,i_2} [A_3]^{s_3}_{i_2,i_3} \cdots [A_n]^{s_n}_{i_{n-1}} |s_1,\ldots,s_n\rangle$$

To keep the matrix problems arising in DMRG, one enforces

$$\sum_{s_k} [A_k^\dagger]^{s_k} [A_k]^{s_k} = I$$

this is nothing more than condition that A is valid quantum superoperator!

DMRG works because it is simulating D dimensional quantum circuit!

Conclusion

- MPS Tomography and Symmetrized Tomography provide tractable methods for analyzing MUSIQC architecture.
- Will produce toolkit that describes measurement protocol and processes measurement data to execute this tomography
- Quantum algorithms - System optimization will guide, but also design quantum circuits to foil best classical algorithms and our tomography